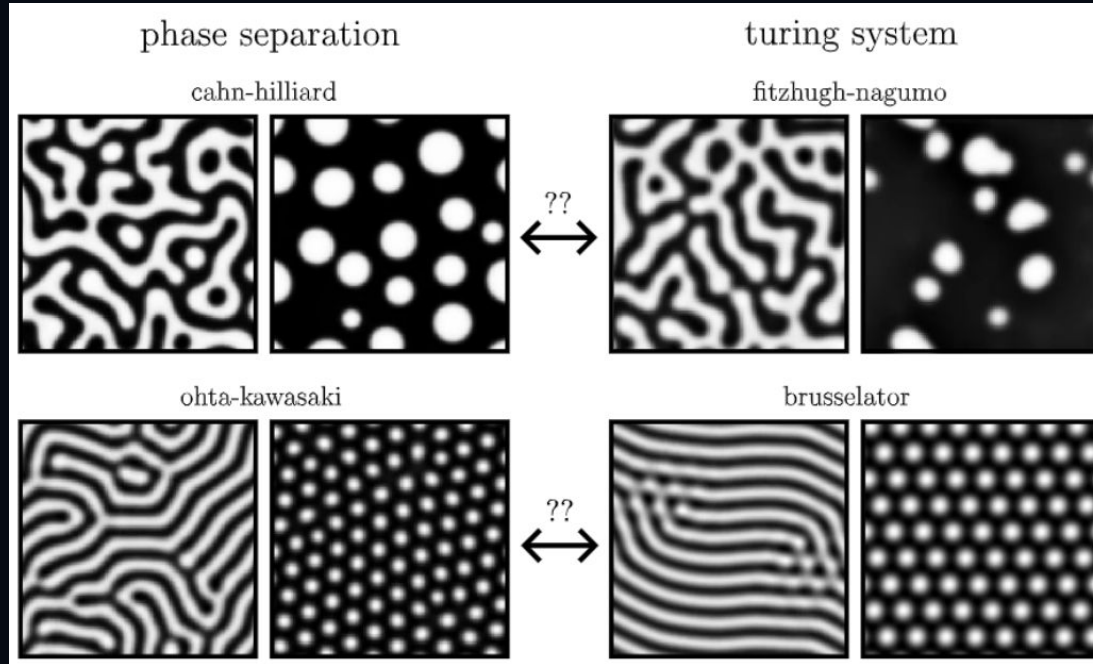


# Turing Patterns and Phase Separations:

## Ethan Coleman & Zane Perry

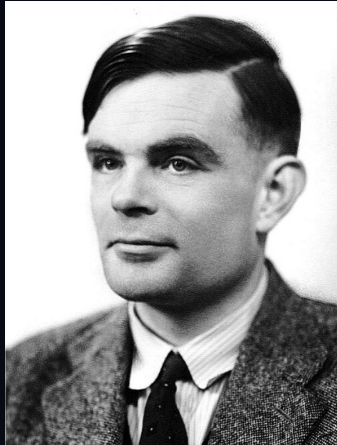


# Motivation



# The Turing Instability

Alan Turing



Proposed diffusion-driven chemical instabilities as the “Chemical Basis of Morphogenesis” in 1952

Ilya Prigogine



Developed the “Brusselator model”, a classical turing system and won the nobel prize in chemistry in 1977



# Derivation of Turing Systems

Balance Laws:

$$\frac{\partial}{\partial t} \int_V c(\mathbf{x}, t) dV = - \int_S \mathbf{J} \cdot d\mathbf{S} + \int_V f dV$$

Divergence Theorem and Ficks Law:

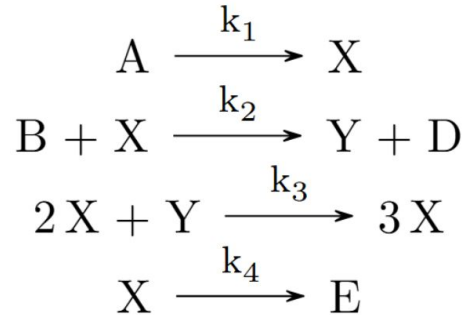
$$\frac{\partial c}{\partial t} = f + D \nabla^2 c$$

$$\begin{aligned} \frac{\partial X}{\partial t} &= F(X, Y) + D_X \nabla^2 X \\ \frac{\partial Y}{\partial t} &= G(X, Y) + D_Y \nabla^2 Y \end{aligned}$$

- Reaction Kinetics (Standard nonlinear dynamics)
- Diffusion Terms (Spatial PDE Extensions)



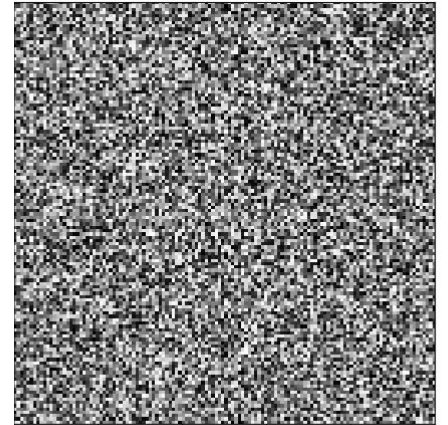
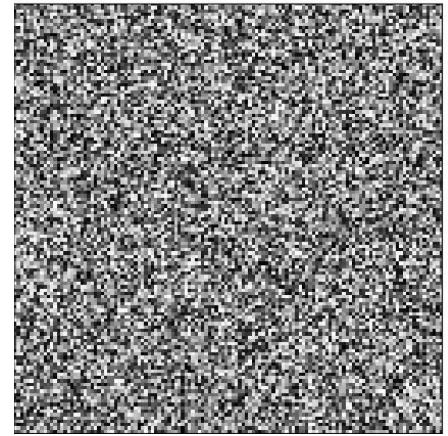
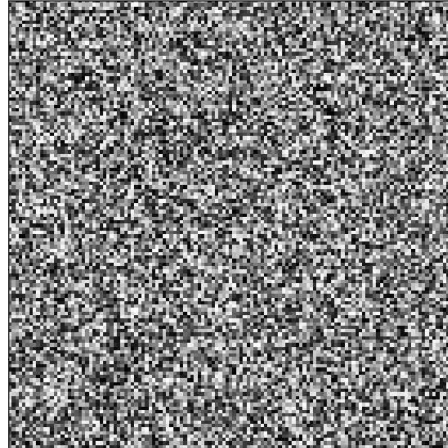
# Brusselator Model



Law of Mass Action:

$$\partial_t X = A - (B + 1)X + X^2 Y + \nabla^2 X$$

$$\partial_t Y = BX - X^2 Y + D \nabla^2 Y$$



# Derivation of Cahn-Hilliard and Ohta-Kawasaki

Conservation:

$$\frac{\partial \phi}{\partial t} + \nabla \cdot \mathbf{J} = 0$$

Chemical Potential:

$$\partial_t \phi = M \nabla^2 \mu_{\text{tot}}$$

Free Energy:

$$F[\phi] = \int_{\Omega} \left( f_0(\phi) + \frac{\kappa}{2} |\nabla \phi|^2 \right) dx + \frac{\alpha}{2} \int_{\Omega} \int_{\Omega} G(x-y) (\phi(x) - \phi_0) (\phi(y) - \phi_0) dx dy$$

- Cahn-Hilliard (Local Terms, Double Well & Interfaces)
- Ohta-Kawasaki (Long Range Separation Penalty)

$$\mu_{\text{tot}}(x) = \mu(\phi(x)) - \kappa \nabla^2 \phi(x) + \alpha \int_{\Omega} G(x-y) (\phi(y) - \phi_0) dy.$$

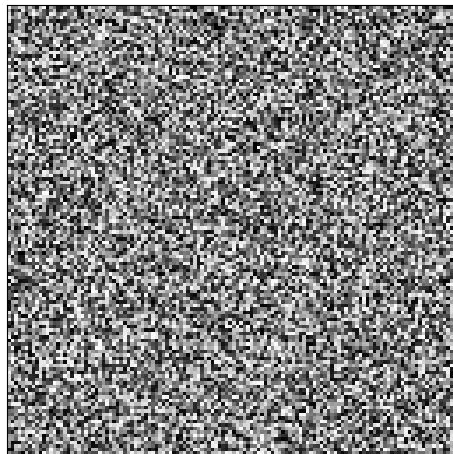
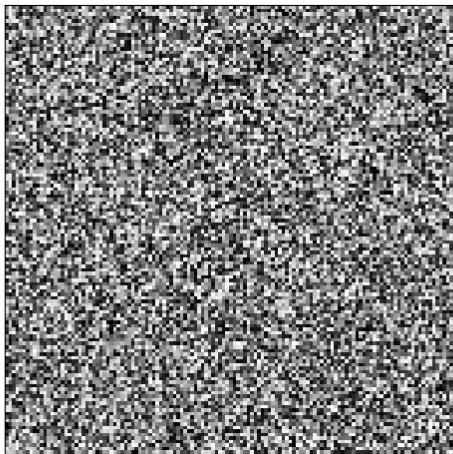




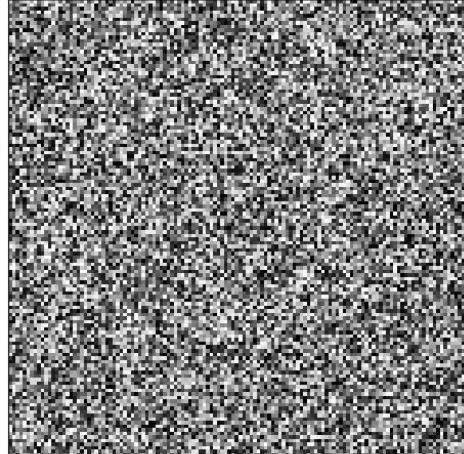
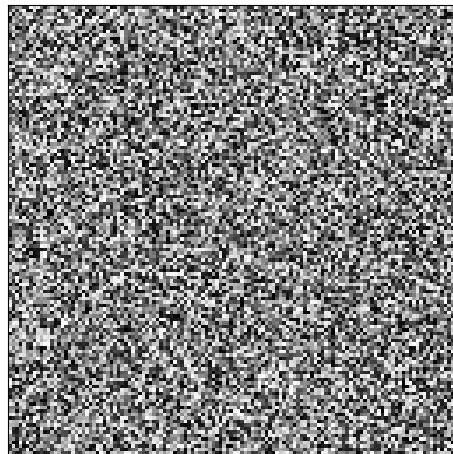
# Ohta-Kawasaki Model

$$\partial_t \phi = M \nabla^2 (\mu(\phi) - \kappa \nabla^2 \phi) - \alpha (\phi - \phi_0)$$

Cahn-Hilliard



Ohta-Kawasaki



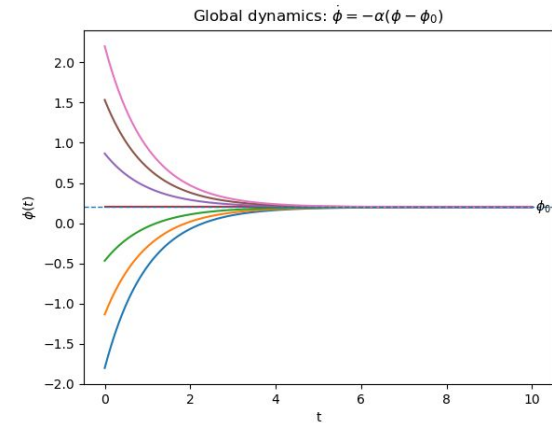
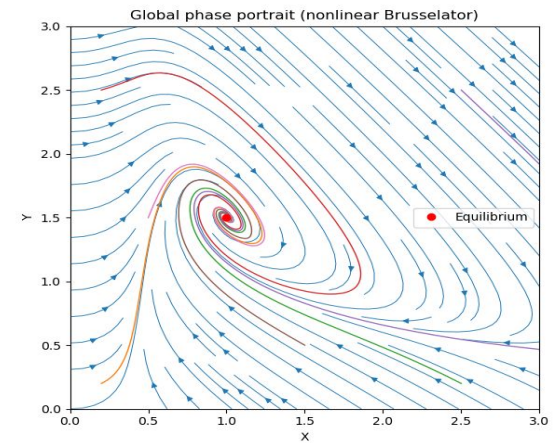
# Linear Phase Fields

Brusselator:

$$\begin{bmatrix} \partial_t X \\ \partial_t Y \end{bmatrix} = \begin{bmatrix} B-1 & A^2 \\ -B & -A^2 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

Ohta-Kawasaki:

$$\dot{\phi} = -\alpha(\phi - \phi_0)$$





# Dispersion Relations

Fourier Eigenmodes:

$$e^{\lambda t + i k x}$$

Brusselator:

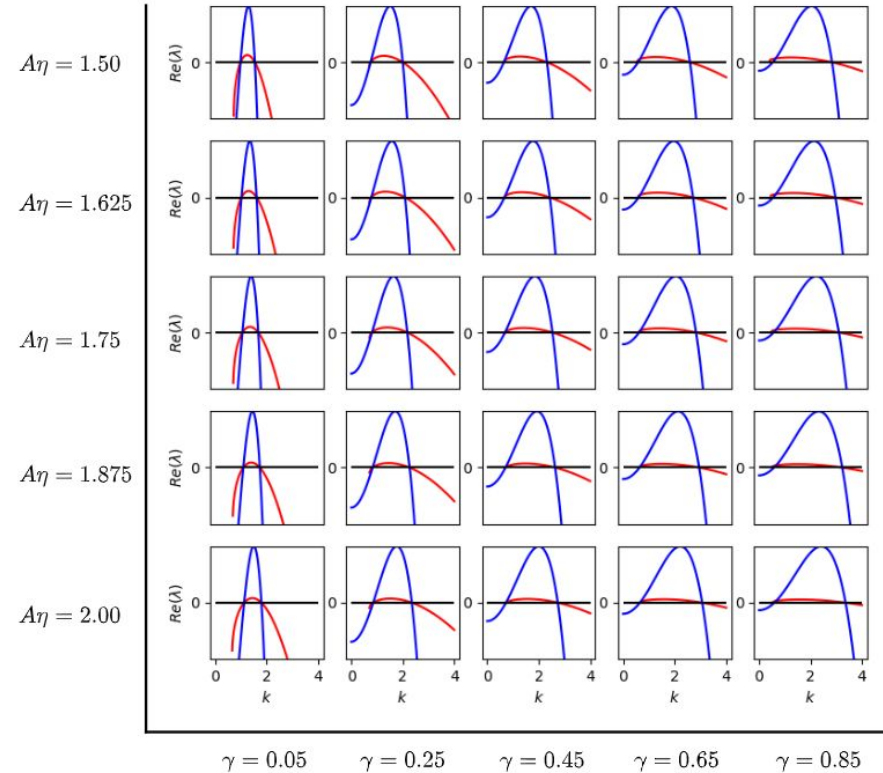
$$\lambda(k) = \frac{\tau(k) \pm \sqrt{\tau(k)^2 - 4\Delta(k)}}{2}$$

Ohta-Kawasaki:

$$\lambda(k) = -M\mu'(\phi_0)k^2 - M\kappa k^4 - \alpha$$

Dispersion Relations

— Ohta-Kawasaki  
— Brusselator



# Critical Wavenumbers

Brusselator:

$$k_c = \left( \frac{D(B-1) - A^2 \pm \sqrt{(A^2 - D(B-1))^2 - 4DA^2}}{2D} \right)^{1/2}$$

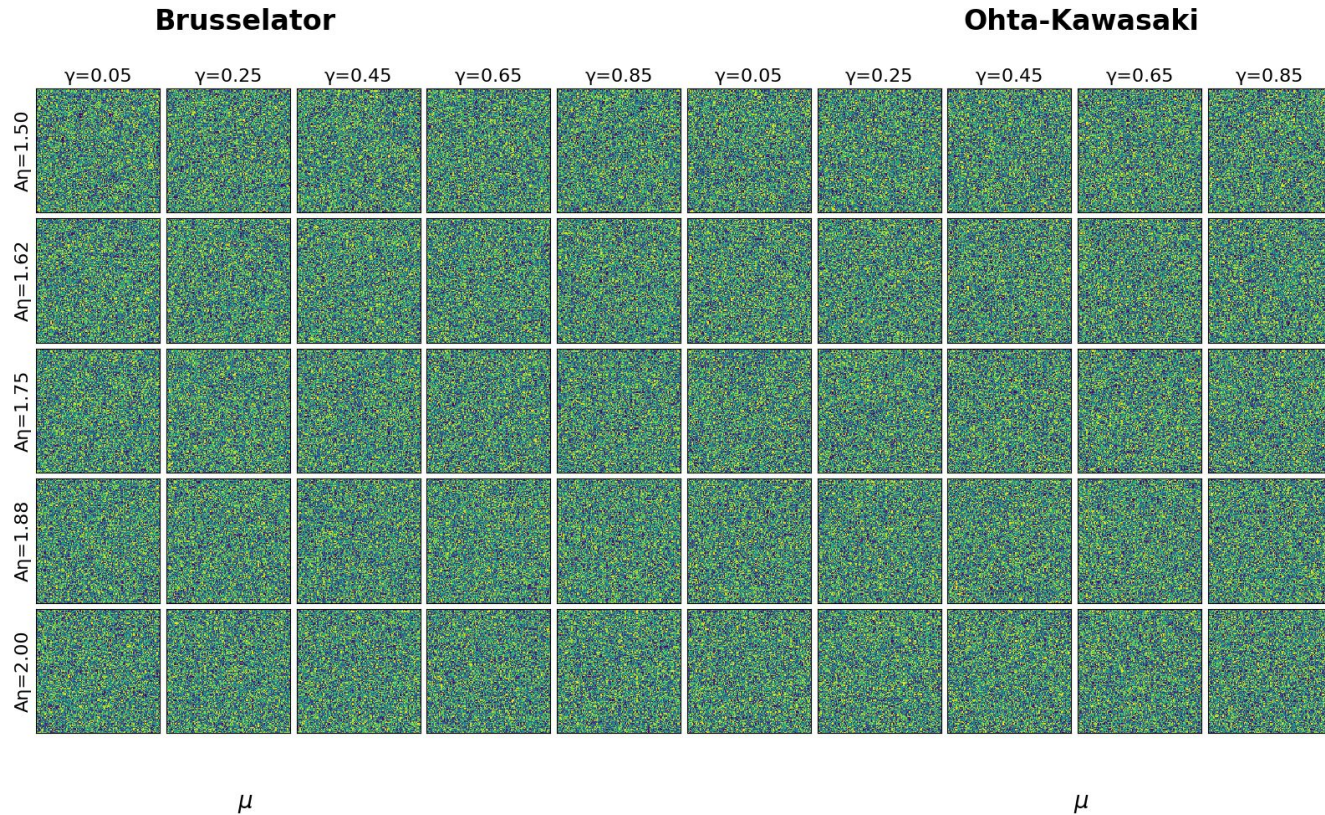
Ohta-Kawasaki:

$$k_c = \left( \frac{-\mu'(\phi_0) \pm \sqrt{\mu'(\phi_0)^2 - 4\kappa\alpha}}{2\kappa} \right)^{1/2}$$

The relationships:  $\kappa = D, \alpha = A^2$ , and  $\partial_\phi \mu(\phi_0) = A^2 - D(B-1)$



# Numerical Results

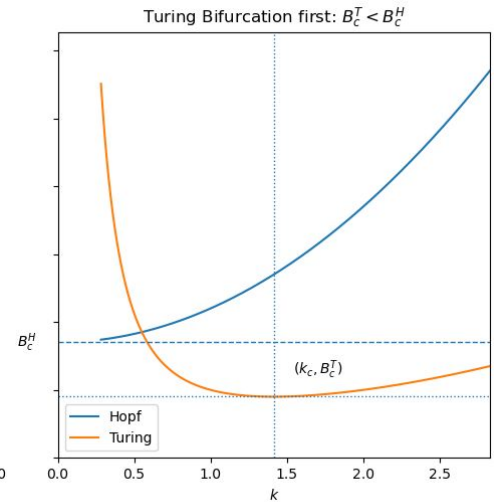
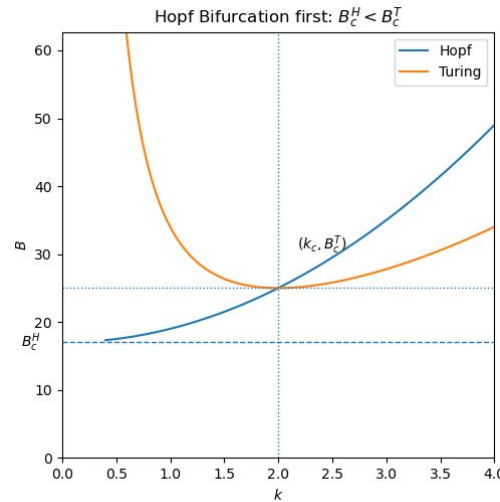


# Bifurcation Analysis

$$B_H = 1 + A^2 \quad (B_T, k_c) = ((1 + A\eta)^2, \sqrt{A\eta}), \quad \eta \equiv \sqrt{\frac{1}{D}}$$

$$\lambda(k) = \frac{\tau(k) \pm \sqrt{\tau(k)^2 - 4\Delta(k)}}{2}$$

$$0 < \eta < \frac{\sqrt{1 + A^2} - 1}{A}$$



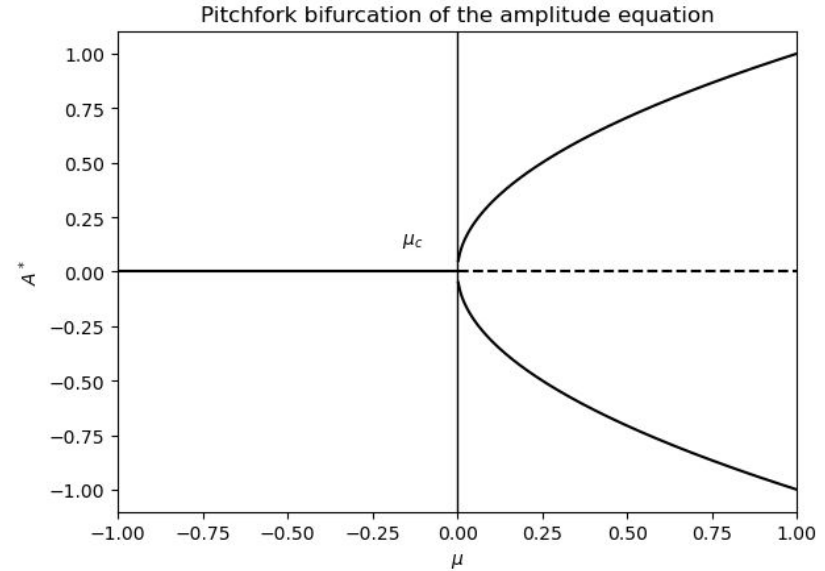
# Amplitude Equations

Critical Mode Expansion:

$$\mathbf{u}(\mathbf{r}, t) = \sum_{j=1}^N \mathbf{u}_0(A_j(\mathbf{r}, t)e^{i\mathbf{k}_j \cdot \mathbf{r}} + \text{c.c.}) + \dots$$

Newell-Whitehead-Segel Equation:

$$\partial_t A = \mu A - g|A|^2 A + \varepsilon_0^2 \left( \partial_x + \frac{1}{2ik_c} \partial_y^2 \right)^2 A$$





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